

FREDERICK, OK, USA

WMO: 723528

Lat: 34.344N

Lon: 98.983W

Elev: 1255

StdP: 14.04

Time Zone: -6.00 (NAC)

Period: 98-19

WBAN: 03981

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.0	21.7	3.3	6.8	25.0	8.8	9.0	31.3	32.4	46.0	28.6	46.2	10.3	10	0.587

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.4	104.3	72.7	101.6	73.1	99.3	73.0	78.2	93.9	77.0	92.2	75.9	90.6	12.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.0	132.7	83.4	72.9	128.0	82.0	72.2	124.8	81.6	42.5	94.6	41.2	92.1	40.2	90.4	90.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.0	25.0	22.5	11.2	107.1	4.1	3.4	8.2	109.5	5.9	111.5	3.6	113.4	0.6	115.9
			9.6	80.4	3.9	3.7	6.8	83.1	4.5	85.2	2.4	87.3	-0.4	90.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.0	42.2	45.4	54.7	63.1	72.3	81.6	85.1	84.7	76.8	64.8	53.1	42.8	(6)
	DBStd	17.52	9.41	10.57	9.79	8.27	7.91	5.43	4.81	5.05	6.65	8.69	9.20	9.08	(7)
	HDD50	871	273	192	63	9	0	0	0	0	0	7	71	256	(8)
	HDD65	2923	707	550	340	128	27	0	0	0	4	111	366	690	(9)
	CDD50	5977	31	64	209	401	693	947	1088	1075	805	467	165	32	(10)
	CDD65	2554	0	2	21	71	255	497	623	610	359	106	10	0	(11)
	CDH74	31925	16	56	279	860	2792	6186	8534	8038	3907	1114	134	9	(12)
	CDH80	17538	0	11	68	296	1317	3468	5142	4813	1986	410	26	1	(13)
(14) Wind	WSAvg	11.2	10.8	11.6	12.4	13.0	12.2	12.2	10.1	9.6	9.9	10.9	10.9	10.5	(14)
(15) Precipitation	PrecAvg	28.1	0.9	1.2	1.9	2.1	4.3	3.5	2.0	2.7	3.6	2.9	1.5	1.1	(15)
	PrecMax	44.8	4.4	4.1	5.6	5.4	11.7	10.6	4.9	7.9	8.4	11.3	5.2	5.2	(16)
	PrecMin	18.4	0.0	0.1	0.1	0.0	0.3	0.5	0.0	0.0	0.0	0.1	0.0	0.0	(17)
	PrecStd	6.4	1.0	1.0	1.4	1.5	2.9	2.4	1.4	2.1	2.4	2.8	1.3	1.1	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.2	81.2	87.3	93.2	100.6	106.1	107.7	108.1	102.5	92.9	83.6	74.0	(19)
		MCWB	53.9	58.8	61.5	63.6	69.6	72.6	71.8	72.3	70.9	68.7	63.7	54.8	(20)
	2%	DB	69.7	73.9	81.0	87.9	96.4	101.3	103.9	104.1	98.4	88.7	77.9	67.6	(21)
		MCWB	51.9	56.3	61.8	65.2	70.5	73.1	73.1	72.9	72.3	67.9	60.8	53.6	(22)
	5%	DB	63.7	68.5	76.6	82.4	91.4	98.6	101.2	101.2	94.2	84.3	73.0	62.9	(23)
		MCWB	48.9	53.9	60.5	64.0	69.9	73.2	73.6	73.2	71.8	66.3	58.7	51.1	(24)
	10%	DB	58.1	63.3	72.0	79.0	87.9	95.1	99.0	98.9	90.7	80.9	68.3	57.4	(25)
		MCWB	46.9	51.3	58.0	62.9	69.3	73.6	73.8	73.0	71.1	64.9	57.3	48.6	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.7	63.9	68.7	73.0	76.8	80.5	80.8	78.3	77.4	74.7	67.7	62.3	(27)
		MCDB	69.7	72.9	79.7	81.2	90.9	96.5	96.8	94.6	91.4	85.3	74.9	67.1	(28)
	2%	WB	54.5	59.8	65.9	70.0	74.7	77.7	78.3	77.0	75.7	72.3	64.2	57.0	(29)
		MCDB	64.6	69.3	75.6	80.2	87.5	92.8	94.1	93.5	88.7	82.8	72.0	63.0	(30)
	5%	WB	50.8	56.1	63.5	68.0	73.2	76.3	77.1	75.9	74.3	69.9	61.8	52.6	(31)
		MCDB	60.8	66.0	71.3	77.5	85.5	90.8	92.4	92.0	86.6	78.9	70.0	60.0	(32)
	10%	WB	47.5	52.3	60.2	66.1	71.5	75.1	75.8	74.9	73.1	67.7	59.0	48.6	(33)
		MCDB	58.4	61.9	68.9	74.6	83.0	89.1	90.7	90.7	84.9	76.3	67.0	57.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	24.2	23.6	24.5	24.7	23.3	23.1	23.4	23.8	23.2	23.6	23.5	21.8	(35)
		MCDBR	33.9	33.4	31.3	31.1	28.9	27.6	27.6	27.8	27.4	29.0	29.7	30.3	(36)
	5% WB	MCWBR	19.7	18.4	16.5	15.4	10.9	8.7	7.6	8.1	9.1	12.9	15.7	18.3	(37)
		MCWBR	30.5	29.2	25.2	24.5	24.5	23.6	22.8	23.3	22.1	22.7	24.5	25.4	(38)
(40) Clear-Sky Solar Irradiance	taub		0.287	0.299	0.329	0.366	0.398	0.420	0.419	0.413	0.381	0.347	0.312	0.289	(40)
		taud	2.530	2.504	2.439	2.357	2.340	2.311	2.358	2.319	2.382	2.465	2.497	2.522	(41)
	Ebn at Noon		296	303	300	292	282	274	274	274	279	281	283	288	(42)
		Edh at Noon	26	30	34	39	40	41	39	40	36	30	26	24	(43)
(44) All-Sky Solar Radiation	RadAvg		961	1151	1534	1865	2006	2211	2209	2018	1679	1298	1006	819	(44)
	RadStd		97	159	96	147	209	186	130	117	123	178	125	86	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page